SPG-OD: Spectral-Prior-Guided Object Detection for Hyperspectral Remote Sensing Images

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***Abstract*—With the rapid increase in hyperspectral images (HSIs), hyperspectral object-level detection (HOD) has become an important task. However, existing methods often neglect the material similarity between the objects and the background, which increases the difficulty of object–background discrimination. Meanwhile, they have not sufficiently exploited the intrinsic advantage of HSIs, namely their continuous full-spectrum representation, which jointly captures global spectral context, local spectral variations, and dependencies across bands. To address these limitations, this article proposes the Spectral-Prior-Guided Object Detection (SPG-OD) method by integrating spectral prior information. We introduce two key modules: the grouped local spectral enhancement module (GLSEM), which enhances local spectral variations to improve object–background discrimination when objects and background share similar materials, and the spectral objectness prior module (SOPM), which uses prior spectral curves (PSCs) to guide detection. The experimental results on three large-scale HOD datasets show that SPG-OD achieves competitive overall performance and demonstrates the effectiveness of incorporating spectral prior information for hyperspectral object detection.**

***Index Terms*—Hyperspectral image (HSI), object detection, spectral prior information.**

I. Introduction

HYPERSPECTRAL images (HSIs) record hundreds of narrow and contiguous spectral bands, providing rich spectral–spatial information [1], [2], [3] beyond conventional RGB images [4]. Different from multispectral or RGB images [5] that only capture limited spectral responses, HSIs can characterize the detailed reflectance variations [6] of ground objects across a wide wavelength range. Such dense spectral signatures make it possible to distinguish materials with similar spatial appearances [7] but different physical or chemical properties. Owing to this capability, HSIs have been widely used in remote sensing applications [8], [9], [10], [11], including land-cover classification [12], [13], [14], mineral exploration, vegetation monitoring, precision agriculture, urban mapping, maritime surveillance, and environmental assessment.

Meanwhile, recent remote sensing studies have shown that advanced representation learning frameworks, such as Mamba-based models, spectral angle mapping (SAM)-prior guidance, and prototype-based prompt tuning, can effectively improve change detection, change captioning, and multimodal classification tasks [15], [16], [17], [18]. These studies further motivate the exploration of more effective spectral–spatial representation learning for HSI object-level detection.

HSI object detection focuses on identifying objects within HSI using spectral and spatial features. This process is divided into hyperspectral target detection (HTD) [19], [20] and hyperspectral anomaly detection (HAD), depending on whether the prior spectral curve (PSC) is available. HAD aims to detect a limited number of objects that deviate from the surrounding environment when the PSC is not known [21], [22], [23]. In contrast, HTD is designed to locate and classify objects whose spectral curves are similar to the PSC. However, the aforementioned methods, being primarily focused on pixel-level analysis, tend to give limited consideration to spatial features, making it difficult for them to achieve object-level localization of targets. To address this limitation, hyperspectral object detectors were subsequently proposed [24], [25], [26], [27].

Unlike supervised pixel-level detection methods, hyperspectral object-level detection (HOD) [28] does not require PSCs of the object of interest in the inference stage. Instead, they are trained on large-scale HSIs to learn both the spectral and spatial features of the object. By jointly exploiting spectral–spatial features, these methods enable object-level localization and classification of the objects of interest.

Recent HOD studies have attempted to adapt deep object detectors to HSIs. Yan et al. [24] introduce a CNN-based object-level detection model for spectral feature extraction and end-to-end training. He et al. [26] propose S2A and SDCM to address spectral redundancy and interband heterogeneity through band selection, dimensionality reduction, and cross-modal feature modeling. Jang et al. [29] further explore HOD for hyperspectral remote sensing images by combining spatial features. Despite these advances, existing methods still

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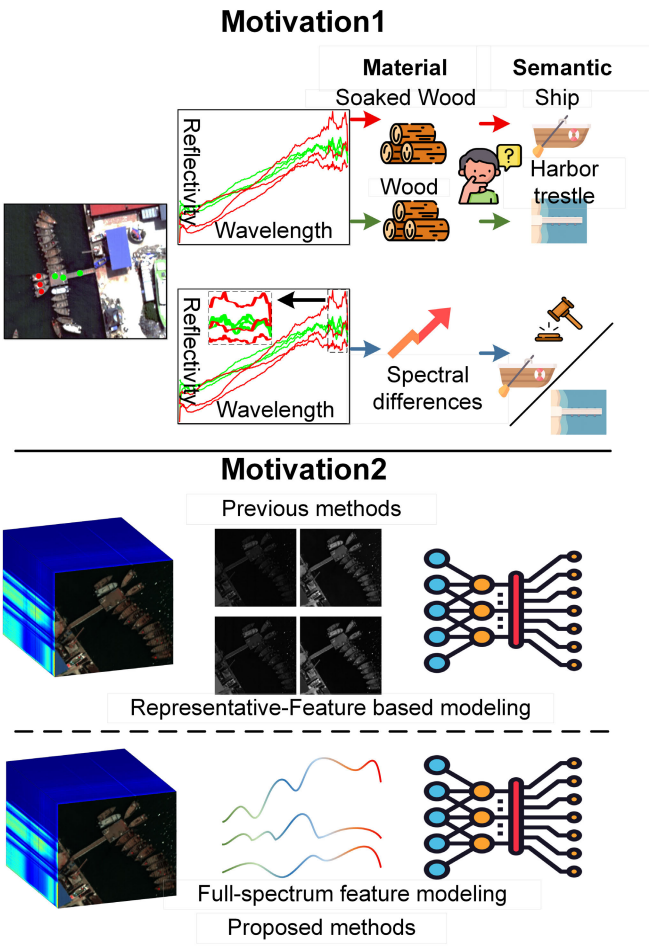


Fig. 1. Top: Motivation 1: Objects made of the same material have similar global spectral curves but may differ in semantics. For example, wooden boats and wooden harbor trestle, although made of the same material, show subtle local spectral differences due to the boat being submerged in water and the drier surface of the wooden harbor trestle. Bottom: Motivation 2: Previous HOD methods relied on band selection or dimensionality reduction, which may discard full-spectrum information.

insufficiently exploit two important characteristics of HSIs, which can be summarized from the following two aspects.

Aspect 1: While spectral curves of the same material generally show similar overall trends, they can convey different semantic information, as shown in Fig. 1. For example, a wooden harbor trestle and a small wooden boat, both made of wood, represent different semantics: the boat as an object and the pier bridge as background.

The spectral curves reveal similar overall trends but subtle local differences, as seen in the magnified view in Fig. 1. Specifically, the submerged wooden boat and the dry pier bridge exhibit sharper peaks in certain spectral regions due to environmental factors. To address this, we propose the grouped local spectral enhancement module (GLSEM), which enhances discriminative local spectral cues to improve object–background discrimination.

Aspect 2: Existing HOD methods [26], [27], [29] usually rely on band selection [30] or dimensionality reduction, which preserve only partial spectral responses and may weaken the continuous spectral structure of HSIs, as shown in Fig. 1. As a result, the global spectral context, local spectral variations, and cross-band dependencies cannot be sufficiently exploited for object detection. To address this issue, the spectral objectness prior module (SOPM) transforms spectral prior information into soft objectness guidance, enabling the detector to emphasize object-related regions in a full-spectrum manner.

The main contributions of this article are summarized as follows.

1) A spectral-prior-guided HOD framework, termed SPG-OD, is proposed to improve HOD by jointly modeling local spectral discrimination and prior-guided objectness.
2) A GLSEM is developed to amplify discriminative local spectral variations within a unified full-range feature space, thereby improving object–background discrimination when similar materials are shared.
3) A SOPM is proposed to transform spectral prior information into soft objectness supervision, enabling the network to implicitly learn object-related spectral responses across different HSIs.
4) Extensive experiments on three large-scale HOD datasets demonstrate that the proposed method achieves competitive overall performance, with clear gains on small and medium-sized objects.

The remainder of this article is organized as follows. Section II reviews the related literature. Section III describes the proposed method, including GLSEM and SOPM. Section IV presents the experimental results and corresponding analysis. Finally, Section V concludes this article.

II. Related Work

Due to the limited number of existing HOD methods, this section mainly reviews supervised pixel-level detection methods for HSI and RGB object detection methods for conventional visible-light images.

A. Supervised Pixel-Level Detection Methods for Hyperspectral Imagery

Supervised pixel-level detection methods for HSIs, commonly referred to as HTD methods, aim to identify pixels whose spectra match given PSCs. Early HTD methods are mainly based on classical signal processing techniques, such as constrained energy minimization, spectral match filter (SMF), adaptive cosine estimator (ACE), generalized likelihood ratio test (GLRT), and orthogonal subspace projection (OSP), which rely on PSCs to suppress background responses.

With the progress of deep learning, the HTD field has witnessed the emergence of numerous innovative methods. Due to the extremely scarce availability of the PSC, current deep learning approaches for HTD mainly follow two paradigms: transfer-learning-based and sample-generating-based methods.

For sample-generating-based techniques, Li et al. [31] propose a method where pixel pairs are generated by permuting and combining limited samples, with pairs from the same category assigned a value of 0 and those from different categories assigned a value of 1. Du and Li [32] improve this by introducing subtraction pixel-pair features, transforming the HTD task into a classification task [33]. Zhang et al. [34] develop HTD-Net, where target samples are generated using an autoencoder, and background samples are filtered through linear prediction. More recently, TSCNTD [35] and STTD

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[36] expand sample pairs by combining confident background samples and PSC. Moreover, graph attention and Transformer-based structures have also been explored to enhance spectral representation learning for HTD, such as the GATformer-based progressive triplet network [37].

For transfer-learning-based methods, samples from additional domains are incorporated into the target domain to address the limited availability of PSC. In 2020, Feng et al. [38] proposed an HTD method that combines spectral and spatial information, transferring pixel pairs from the source domain to the target domain to overcome sample insufficiency. Shi et al. [39] introduced a semi-supervised domain-adaptive few-shot learning (SDAFL) model, which transfers pretrained similarity measures through adversarial learning.

Although the aforementioned methods perform well, they largely overlook spatial features. To address this, the semantic-transferred prior for HTD with spatial–spectral association (SSAD) [40] and the transfer-learning-based TLH2TD [41] are proposed, both transferring spatial semantic features from RGB images to HSIs. However, due to the limited luminous flux in HSIs, which is spread across more bands, the spatial features differ significantly from those of RGB images, leading to poor generalization.

B. RGB Object Detection Methods for Conventional Visible-Light Images

Deep-learning-based RGB object detection methods are divided into two categories: two-stage and single-stage detectors. Two-stage detectors use a coarse-to-fine approach, from region proposal network (RPN) to region of interest (RoI), while single-stage detectors directly predict object classes and bounding box coordinates in a single pass without the region proposal step.

Faster R-CNN [42], the first real-time object detector, introduced a two-stage approach. Later, the feature pyramid network (FPN) improved multiscale object detection by considering semantic features at various levels. However, despite superior accuracy, two-stage models face high computational complexity, limiting their deployment on mobile devices.

On the other hand, the YOLO framework [43] represents single-stage detectors, which parallelly predict different image regions. YOLOv5 introduces a modular PyTorch-based design, followed by YOLOv6, which balances computation efficiency and accuracy with an EfficientRep backbone. YOLOv7 refines real-time performance using E-ELAN and reparameterized convolution. YOLOv8 employs an anchor-free mechanism to improve accuracy and efficiency, while YOLOv9 tackles information bottlenecks with enhanced feature propagation. YOLOv10 introduces a dual-head strategy to stabilize predictions, and YOLO11 further improves computational efficiency with parallel spatial attention and smaller convolution.

Although these methods perform well, most rely on non-maximum suppression (NMS) postprocessing, which cannot be optimized during training and fails to create an end-to-end network. To address this, DETR uses a transformer-based approach, eliminating the need for NMS by directly optimizing the correspondence between predicted and ground-truth bounding boxes. However, its high computational cost limits real-time applicability. RT-DETR [44] solves this by efficiently processing multiscale features with a hybrid encoder and improving accuracy through refined query embeddings. RT-DETRv2 [45] further enhances generalization by selectively extracting multiscale features in the decoder with a deformable attention mechanism.

These RGB-based object detection methods have demonstrated remarkable performance on public RGB datasets. However, since they are not specifically designed for HSI and fail to account for spectral features, directly applying them to HOD tasks often results in significant performance degradation.

Subsequently, specialized HOD methods were developed. Double-FPN [29] represents the first attempt at HOD. However, it utilizes principal component analysis (PCA) to compress spectral features into a smaller subspace, resulting in suboptimal detection performance. Later, He et al. [27] proposed the S2A [26] and SDCM algorithms, which aim to address the challenges of spatial–spectral feature complementarity and spectral heterogeneity. However, both methods rely on existing band selection techniques [46] and do not fully leverage spectral features. In addition, they fail to account for object–background confusion caused by similar materials. These limitations motivate us to design a spectral-prior-guided framework that preserves full-spectrum information while enhancing local spectral cues for object–background discrimination.

III. Methodology

To address the aforementioned issues, we propose the SPG-OD for HSIs. First, HSIs are processed by GLSEM, which enlarges local spectral-feature groups and projects multiple local spectral responses into a unified full-range feature space. This approach emphasizes subtle local differences caused by varying surrounding environments for the same material, thereby improving object–background discrimination. Next, the improved features are fed into the SOPM to learn the prior spectral features of objects across different HSIs. Subsequently, the resulting feature maps are input into the spatial feature processing module to embed spatial feature priors for different objects. Finally, to complete the detection, the features obtained in the previous steps are passed through the detection head.

In essence, the proposed method addresses HOD from two complementary perspectives. GLSEM enhances subtle local spectral differences that are useful for distinguishing objects from backgrounds with similar materials, while SOPM converts spectral prior information into an objectness-like supervision signal that guides the network toward target-related regions.

The overall network architecture of our proposed method is depicted in Fig. 2.

A. Grouped Local Spectral Enhancement Module

The goal of GLSEM is not to simply amplify spectral responses, but to enhance locally discriminative spectral variations while preserving stable object-level semantics. To this end, the input HSI is first projected into a unified feature space and then modulated group by group along the spectral dimension.

We first define the notation used in GLSEM. The HSI is denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$, where H, W, and L represent the height, width, and number of spectral bands, respectively. To

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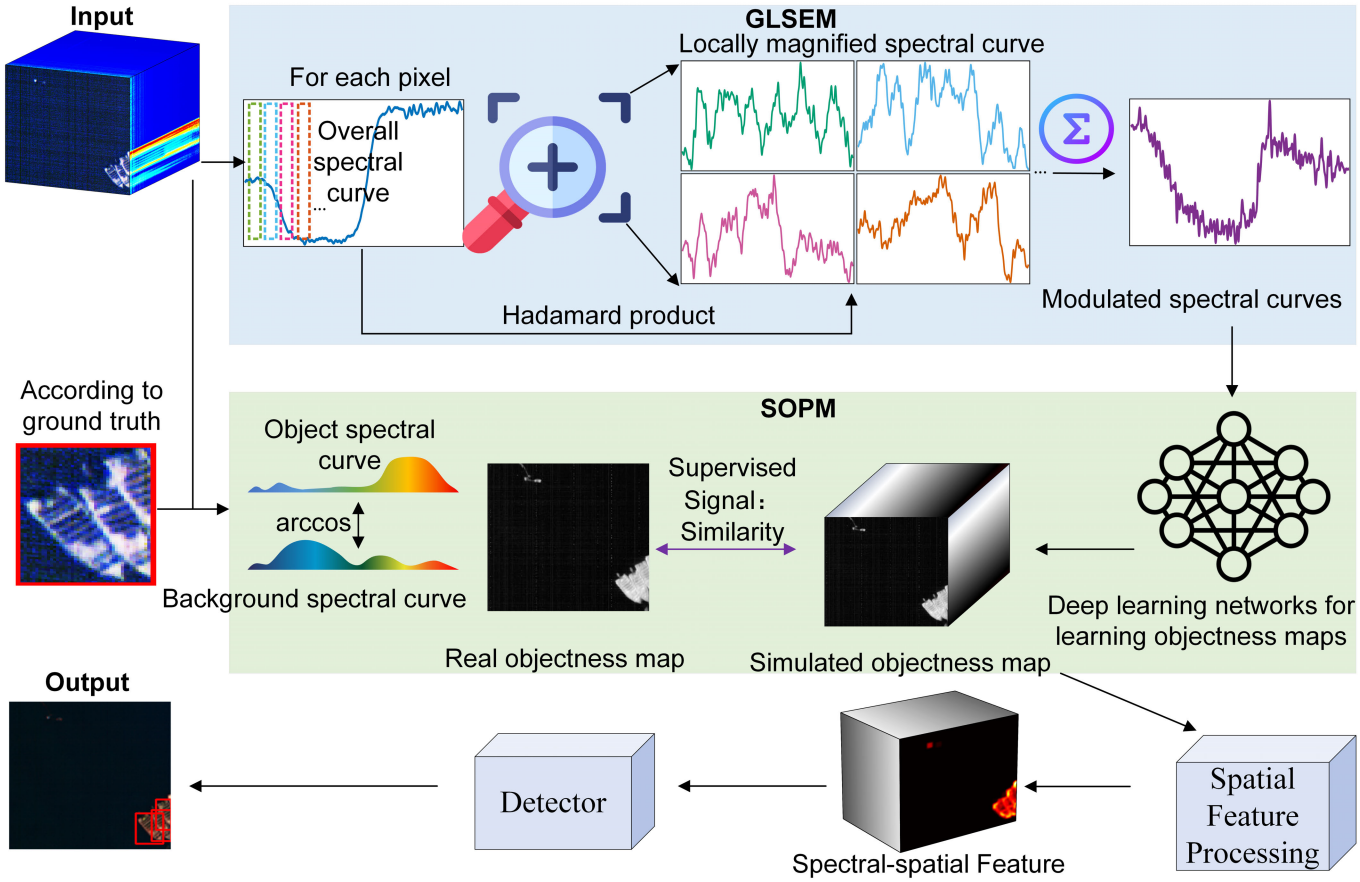


Fig. 2. Overall architecture of the proposed SPG-OD framework. The input HSI is first projected into a unified spectral feature space and processed by the GLSEM, which divides the spectral features into several local groups and enhances discriminative local spectral variations through groupwise modulation. The enhanced feature is then fed into the SOPM. During training, SOPM generates a real spectral objectness map from the PSC and ground-truth boxes, and uses it to supervise a learnable objectness prediction branch. During inference, this branch predicts a simulated objectness map from the enhanced spectral feature to guide object-related feature responses. Finally, the resulting spectral–spatial features are processed by the spatial feature processing module and sent to the detector for object-level localization.

unify inputs with different band numbers, we employ a 2-D frozen convolution as a fixed spectral projection. Compared with a trainable projection, the frozen convolution avoids unstable early spectral mapping and potential overfitting under limited HSI annotations, while the subsequent GLSEM and SOPM modules learn the discriminative spectral representations.

The feature cube is $\mathbf{X}_{F1} \in \mathbb{R}^{H\times W\times L_1}$, and is formulated as

$$f(\mathbf{T})^{(o)}[i,j] = \sum_{c=1}^{C_{\text{in}}}\sum_{u=0}^{k_h-1}\sum_{v=0}^{k_w-1} \mathbf{T}^{(c)}\left[i\cdot s+u-p,\ j\cdot s+v-p\right]\cdot \mathbf{W}_{c,u,v}^{(o)} \quad (1)$$

$$\mathbf{X}_{F1}^{(o)}[i,j] = \left(f_1(\mathbf{X})\right)_{\text{Frozen}}^{(o)}[i,j] \quad \forall o\in\{1,\ldots,L_1\}\quad \forall i\in\{1,\ldots,H\}\quad \forall j\in\{1,\ldots,W\} \quad (2)$$

where C_{in} denotes the number of input channels of $\mathbf{T}$, and k_h and k_w denote the kernel height and width, respectively. The index o indicates the output channel (i.e., the convolution kernel), and $\mathbf{W}$ denotes the corresponding convolution parameters. Moreover, p and s represent the padding size and stride, respectively.

We divide $\mathbf{X}_{F1}$ into P spectral groups along the channel dimension. Let $G = (L_1)/P$ (assuming that L_1 is divisible by P). Then, the uth group is defined as

$$\mathbf{X}_{F1}^{u} = \mathbf{X}_{F1}(:,:,\ (u-1)\cdot G+1 : u\cdot G), \quad u=1,2,\ldots,P \quad (3)$$

where $\mathbf{X}_{F1}^{u} \in \mathbb{R}^{H\times W\times G}$.

In our experiments, we set $L_1 = 256$ since existing large-scale remote sensing HOD datasets typically contain no more than 300 spectral bands. Meanwhile, the choice of P is critical for balancing material and semantic information. We empirically determine its optimal value via experiments reported in Section IV:

$$\mathbf{X}_u = \partial\left(\delta\left(f_u\left(\mathbf{X}_{F1}^{u}\right)\right)\right), \quad u=1,2,\ldots,P \quad (4)$$

where $\delta(\cdot)$ denotes the ReLU nonlinear activation, $\partial(\cdot)$ denotes GroupNorm for feature normalization, and $u \in \{1,2,\ldots,P\}$ indexes the spectral group. In practice, the convolution block $f_u(\cdot)$ is configured to output L_1 channels, yielding $\mathbf{X}_u \in \mathbb{R}^{H\times W\times L_1}$, so that it is channel-aligned with $\mathbf{X}_{F1}$ and can be multiplied elementwise without any broadcasting or additional projection.

Subsequently, the feature cube $\mathbf{X}_{F1}$ is reweighted by each groupwise response map $\mathbf{X}_u$ via elementwise multiplication,

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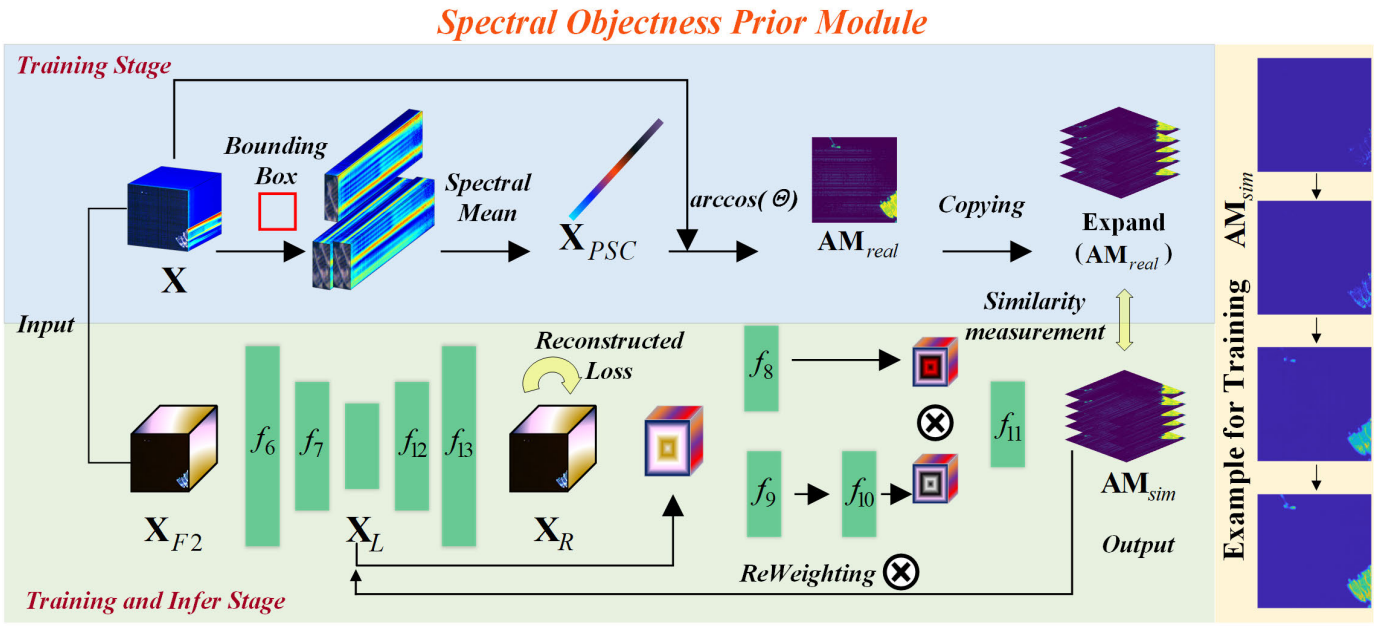


Fig. 3. Overall architecture of the SOPM.

and the P reweighted features are summed to obtain the enhanced output $\mathbf{X}_{F2} \in \mathbb{R}^{H\times W\times L_1}$

$$\mathbf{X}_{F2} = \sum_{u=1}^{P} \mathbf{X}_u \otimes \mathbf{X}_{F1} \tag{5}$$

where $\otimes$ denotes elementwise multiplication. This operation injects group-specific local spectral variations into the unified feature space, thereby strengthening material-sensitive cues while preserving the global semantic structure for object-level detection.

B. Spectral Objectness Prior Module

The goal of SOPM is to transform spectral prior information available during training into a latent objectness guidance that remains usable at inference time. Instead of directly using PSC during testing, SOPM learns to approximate a spectral objectness map through self-reconstruction and similarity supervision.

To fully leverage spectral features and embed generalized prior spectral information, the SOPM is introduced. It is designed by simulating the real objectness map generated from the SAM [47]. The flowchart of this module is shown in Fig. 3.

First, the PSC is crucial for HTD and can serve as a valuable training-stage spectral prior for HOD. To extract the PSC, partial HSIs containing objects are segmented from the raw HSI $\mathbf{X}$. This is achieved using the bounding box specified by $(x_{min}, y_{min}, x_{max}, y_{max})$. The PSC is obtained by calculating the mean value across the entire partial HSI. One such PSC, $\mathbf{X}_{PSC,i} \in R^{1\times L}$ derived from a particular partial HSI, can be expressed as follows:

$$\mathbf{X}_{\mathrm{PSC},i} = \frac{1}{(x_{\max} - x_{\min} + 1)(y_{\max} - y_{\min} + 1)} \times \sum_{x=x_{\min}}^{x_{\max}} \sum_{y=y_{\min}}^{y_{\max}} \mathbf{X}(x, y, :). \tag{6}$$

Based on the formula above, the final $\mathbf{X}_{PSC} \in R^{1\times L}$ can be obtained through

$$\mathbf{X}_{PSC} = \frac{1}{M} \sum_{i=1}^{M} \mathbf{X}_{PSC,i} \tag{7}$$

where M represents the number of targets in the HSI. It should be noted that the PSC is extracted from bounding-box annotations rather than pixel-level masks, and thus some background pixels may inevitably be included. This is an inherent limitation of box-level supervision. However, the extracted PSC is used as a soft spectral guidance signal rather than a strict object mask. In practice, background pixels within different boxes usually exhibit diverse and inconsistent spectral patterns, whereas object pixels tend to share more stable spectral characteristics. Therefore, the influence of background pixels can be partially suppressed when learning the objectness map from multiple objects and training samples. Moreover, since the objectness map is jointly optimized with the detection objective, the network is not forced to exactly follow the noisy box-level prior. Thus, SOPM can still provide useful spectral guidance for enhancing object-related regions.

In SOPM, the PSC is not intended to serve as a category prototype. Instead, it is used as a soft objectness prior during training, encouraging the network to emphasize foreground regions whose spectra are similar to the extracted PSC. To generate the real objectness map guided by spectral features, the SAM is integrated. This mechanism serves to emphasize targets with spectral features similar to the PSC in the spectral dimension. By considering each pixel as a high-dimensional spectral vector, the SAM can calculate the similarity between vectors by measuring the corresponding angles. A smaller angle implies a higher degree of spectral similarity, whereas a larger angle indicates greater spectral dissimilarity. The spectral angle θ between two vectors $\mathbf{A} \in R^{1\times L}$ and $\mathbf{B} \in R^{1\times L}$

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is computed as follows:

$$\theta = cos^{-1}\left(\frac{\mathbf{A}\cdot\mathbf{B}^T}{||\mathbf{A}||||\mathbf{B}||}\right) \tag{8}$$

where $\cdot$ and $\|.\|$ denote the dot product and the two-norm, respectively. The similarity between the PSC and each spectral vector in the raw HSI is calculated to obtain the real objectness map $\mathbf{AM}_{real} \in R^{H\times W}$. Notably, in practical applications, the reciprocal of θ is used. This is done to give prominence to the target whose spectral curve is similar to that of the PSC.

However, the real objectness map is reliant on the bounding box, and thus on the PSC, which is not accessible during the test stage. To approximate the real objectness map, we have devised a reweighting strategy based on a self-reconstructing approach to learning the predicted objectness map. Specifically, the data feature cube $\mathbf{X}_{F2}$ output from the GLSEM is utilized for reconstruction. The middle latent variable $\mathbf{X}_L$ within this reconstruction process captures crucial features that emphasize the targets, making it advantageous for simulating objectness maps $\mathbf{AM}_{sim}$. As a result, the latent variable is chosen to simulate the objectness map by convolution f_6 and f_7.

In general, within a reasonable range of spectral resolutions, the target–background separability is relatively stable and is mainly determined by the underlying spectral characteristics. We adopt two strategies to expand the spectral feature channels, thereby improving robustness to spectral noise and background interference. The first way involves directly increasing the number of spectral feature channels using a 2-D 3×3 convolution f_8. The second way is an indirect method. It first employs a 2-D 1×1 convolution f_9, followed by a 2-D 3×3 convolution f_{10}. This sequence enables the model to better adapt to a wider range of spectral feature variations. Subsequently, the elementwise product of the two obtained variables is fed into a 2-D 1×1 convolution f_{11}. This process yields the simulated objectness map $\mathbf{AM}_{sim} \in R^{H\times W\times(L_1)/P}$, where the number of channels in $\mathbf{AM}_{sim}$ is equal to the number of channels in the latent variable.

Although the latent variable $\mathbf{X}_L$ preserves valuable features, spectral features have not been taken into account sufficiently. To address this, we reweight the latent variable by performing an elementwise product with the simulated objectness map $\mathbf{AM}_{sim}$, which is guided by spectral information. Subsequently, the reconstruction $\mathbf{X}_R \in R^{H\times W\times L_1}$ is accomplished through two symmetric convolutions, i.e., f_{12} and f_{13}.

The self-reconstruction process can be achieved by incorporating a mean squared error (mse) loss. To make the simulated objectness map approximate the real objectness map, an mse loss is also introduced between them. Through this approach, the simulated objectness map can implicitly learn the spectral features. The corresponding loss functions are expressed as follows:

$$Loss_{rec} = \|\mathbf{X}_{F2} - \mathbf{X}_R\|_F^2 \tag{9}$$

$$Loss_s = \|\mathbf{AM}_{sim} - expand\,(\mathbf{AM}_{real})\|_F^2 \tag{10}$$

where the $\|.\|_F$ denotes the Frobenius norm. To align the dimensionality of $\mathbf{AM}_{sim}$ with $\mathbf{AM}_{real}$, the number of channels in $\mathbf{AM}_{real}$ is expanded to $(L_1)/P$.

Algorithm 1 Training Procedure of the Proposed SPG-OD Method

Input: HSI $\mathbf{X}$, Boundingbox $(x_{min}, y_{min}, x_{max}, y_{max})$.
Output: The optimal weight of the model.

while *The current iteration index remains lower than the maximum number of iterations.* **do**
- The data feature cube is generated through equations (5).
- The PSC can be derived via equations (7).
- The real spectral attention map is generated according to equation (8).
- The simulated spectral attention map is derived from the designed convolution structure.
- The comprehensive spatial features are constructed in accordance with the FPN.
- The RPN in combination with RoI pooling is employed to generate and refine the candidate bounding boxes.
- Compute the backpropagation gradient and update the model weights using Equation (11).

end

Algorithm 2 Test Procedure of the Proposed SPG-OD Method

Input: HSI $\mathbf{X}$, IoU threshold, score threshold.
Output: The detection map.

1. The data feature cube is generated by (5).
2. The simulated spectral attention map is calculated using the designed convolution structure.
3. The comprehensive spatial features are constructed following the FPN.
4. The RPN and RoI pooling are utilized to generate and refine the candidate bounding boxes.
5. The candidate bounding boxes are overlaid onto the pseudo-color image of HSI in accordance with the NMS. The NMS algorithm incorporates both an IoU threshold and a score threshold.

C. Overall Loss of Proposed Method

To embed spatial features, we apply spatial feature processing to the feature maps output by SOPM. In this article, we introduce the FPN to obtain robust spatial features. Later, in the detection head, we adopt RPN and RoI techniques, where RPN is used to generate candidate regions that may contain targets, and RoI is used to further extract these regions for object classification and precise localization. Therefore, the overall loss function can be expressed as follows:

$$Loss = Loss_{RPN_RoI} + Loss_{rec} + \alpha Loss_s \tag{11}$$

where α denotes the hyperparameter, and its value is set according to the subsequent parameter-determination experiment. $Loss_{RPN_RoI}$ represents the losses from the RPN and RoI operations, respectively.

Aiming to ensure the reproducibility of our method, the involved parameters of the convolution operation are summarized in Table I. To clearly demonstrate the operation of Spectral-Prior-Guided Object Detection (SPG-OD), the corresponding pseudocode for the training process is shown in

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TABLE I
INPUT–OUTPUT DIMENSIONS AND TENSOR SHAPES OF GLSEM AND SOPM

Module	Step	Operation	Input tensor	Output tensor
Notation	–	Batch dimension is omitted. Tensors are written as $H \times W \times C$. Let $G = L_1/P$.	–	–
GLSEM	Input feature	Raw HSI cube	–	$X \in \mathbb{R}^{H\times W\times L}$
GLSEM	Spectral projection	Frozen 3×3 Conv f_1, $L \to L_1$, padding 1, stride 1	$X \in \mathbb{R}^{H\times W\times L}$	$X_{F1} \in \mathbb{R}^{H\times W\times L_1}$
GLSEM	Spectral grouping	Split X_{F1} into P channel groups: $X_{F1}^u = X_{F1}(:,:,(u-1)G+1:uG)$	$X_{F1} \in \mathbb{R}^{H\times W\times L_1}$	$X_{F1}^u \in \mathbb{R}^{H\times W\times G}$
GLSEM	Group-wise response	$X_u = \mathrm{GN}(\mathrm{ReLU}(f_{u+1}(X_{F1}^u)))$; for $P = 4$, this corresponds to f_2–f_5, each 3×3 Conv, $G \to L_1$	$X_{F1}^u \in \mathbb{R}^{H\times W\times G}$	$X_u \in \mathbb{R}^{H\times W\times L_1}$
GLSEM	Spectral modulation	Element-wise multiplication between each group response and the projected feature	$X_u, X_{F1} \in \mathbb{R}^{H\times W\times L_1}$	$\widetilde{X}_u = X_u \odot X_{F1} \in \mathbb{R}^{H\times W\times L_1}$
GLSEM	Group fusion	Sum all modulated group features: $X_{F2} = \sum_{u=1}^{P} \widetilde{X}_u$	$\{\widetilde{X}_u\}_{u=1}^{P}$	$X_{F2} \in \mathbb{R}^{H\times W\times L_1}$
SOPM	Latent feature extraction	f_6: 3×3 Conv, $L_1 \to 2G$; f_7: 3×3 Conv, $2G \to G$	$X_{F2} \in \mathbb{R}^{H\times W\times L_1}$	$X_L \in \mathbb{R}^{H\times W\times G}$
SOPM	PSC generation	Crop object regions by bounding boxes and average spectra over the cropped pixels and M targets	$X \in \mathbb{R}^{H\times W\times L}$	$X_{PSC} \in \mathbb{R}^{1\times L}$
SOPM	Real objectness map	Compute SAM similarity between each pixel spectrum and X_{PSC}	X, X_{PSC}	$AM_{real} \in \mathbb{R}^{H\times W}$
SOPM	Target expansion	Copy AM_{real} along the channel dimension to match AM_{sim}	$AM_{real} \in \mathbb{R}^{H\times W}$	$\mathrm{expand}(AM_{real}) \in \mathbb{R}^{H\times W\times G}$
SOPM	Direct objectness branch	f_8: 3×3 Conv, $G \to 2G$	$X_L \in \mathbb{R}^{H\times W\times G}$	$A_1 \in \mathbb{R}^{H\times W\times 2G}$
SOPM	Indirect objectness branch	f_9: 1×1 Conv, $G \to 2G$; f_{10}: 3×3 Conv, $2G \to 2G$	$X_L \in \mathbb{R}^{H\times W\times G}$	$A_2 \in \mathbb{R}^{H\times W\times 2G}$
SOPM	Simulated objectness map	$AM_{sim} = f_{11}(A_1 \odot A_2)$, where f_{11} is a 1×1 Conv, $2G \to G$	$A_1, A_2 \in \mathbb{R}^{H\times W\times 2G}$	$AM_{sim} \in \mathbb{R}^{H\times W\times G}$
SOPM	Spectral-guided reweighting	Reweight latent features using the learned objectness map	$X_L, AM_{sim} \in \mathbb{R}^{H\times W\times G}$	$X'_L = X_L \odot AM_{sim} \in \mathbb{R}^{H\times W\times G}$
SOPM	Reconstruction	f_{12}: 3×3 Conv, $G \to 2G$; f_{13}: 3×3 Conv, $2G \to L_1$	$X'_L \in \mathbb{R}^{H\times W\times G}$	$X_R \in \mathbb{R}^{H\times W\times L_1}$
SOPM	Training losses	$\mathcal{L}_{rec} = \|X_{F2} - X_R\|_F^2$, $\mathcal{L}_s = \|AM_{sim} - \mathrm{expand}(AM_{real})\|_F^2$	Matched tensor pairs	Scalar losses

Algorithm 1, and the pseudocode for the inference process is shown in Algorithm 2.

It should be noted that the spectral prior in SOPM is used only as a training-stage supervision signal. The PSC X_{PSC} is computed from annotated object regions in the training set, and AM_{real} is generated only for supervising the learning of AM_{sim}. During inference, neither ground-truth bounding boxes nor X_{PSC} or AM_{real} is used. Given a test HSI, SOPM directly predicts AM_{sim} from the learned latent feature X_L. Thus, the spectral prior exists at inference only implicitly in the learned parameters, and no test-time privileged information or annotation is involved.

IV. EXPERIMENTS AND ANALYSIS

In this section, we perform comparative experiments on three existing large-scale remote-sensing HOD datasets. We compare our proposed approach with existing RGB-based methods and representative HOD methods. This is to validate the effectiveness and robustness of our proposed method.

A. Datasets Description

Due to the high cost of acquiring HSIs, large-scale remote sensing HOD datasets are scarce. As far as we know, the only existing large-scale remote sensing HOD datasets are M2SODAI [29], LWP1.0 [48], and the remake version LWP2.0, which are summarized in Table II.

B. Experiment Setup

1) Comparison Methods: Since most existing HSI detection methods are pixel-level, we evaluate our method by comparing it with six leading RGB-based object-level detection methods and two recent HOD techniques. These methods include Faster-RCNN [42], RT-DETRv2 [45], YOLOv5, YOLOv8, YOLOv10, YOLO11 [43], DoubleFPN [29], and the S2A [26] method. Faster-RCNN, a two-stage detector, is an RGB-based approach. RT-DETRv2 is a detector developed from a transformer architecture for detecting objects in RGB images. YOLOv5, YOLOv8, YOLOv10, and YOLO11 are single-stage detectors based on RGB images. DoubleFPN, as the first to try for a remote sensing HOD method, is a two-stage detector built upon Faster-RCNN. S2A, as an HOD method designed for natural HSIs, is a single-stage approach.

Since RGB object detectors are originally designed for three-channel inputs, we adapt them to HSI by modifying the first convolutional layer to accept B spectral bands, while keeping all subsequent layers unchanged. Specifically, the original first-layer kernel of size $3 \times k \times k$ is replaced with a $B \times k \times k$ kernel, and the remaining network architecture and hyperparameters follow the official implementations of the corresponding methods.

SPG-OD is trained using the Adam optimizer with an initial learning rate of 3×10^{-4}, a batch size of 4, and a weight decay of 0. The training schedule followed the iteration-based setting in Detectron2, and each model was trained for 15 000 iterations with 50 warm-up iterations and a warm-up factor of 0.02. The HSIs are loaded from.mat files, converted to channel-first float tensors, and normalized by per-image min–max normalization.

2) Evaluation Metrics: The proposed method's performance is evaluated qualitatively and quantitatively using detection maps, mean average precision (mAP), and heat maps. Detection maps provide a qualitative assessment, while mAP (including mAP_{50}, mAP_{75}, mAP_{small}, mAP_{medium}, and mAP_{large}) offers a quantitative evaluation based on target size and various IoU thresholds (0.5, 0.75, and 0.5–0.95 with a step of 0.05). All metrics are computed with the pycocotools library. Heat maps are used to show the feature-representing capabilities of different modules.

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TABLE II
DETAILED PARAMETERS OF THE M2SODAI, LWP1.0, AND LWP2.0 DATASETS

Dataset	Category	small ($< 32^2$)	medium (32^2–96^2)	large ($> 96^2$)	Total	HSIs	Instances
M2SODAI (127 bands, 400–1000 nm)							
Training	ship	3980	193	1	4174	1007	4174
Test	ship	496	15	0	511	125	511
LWP1.0 (270 bands, 400–1000 nm)							
Training	ship	439	991	360	1790	1339	1790
Test	ship	90	209	12	311	164	311
LWP2.0 (270 bands, 400–1000 nm)							
Training	ship	165	436	120	721		
	car	61	23	0	84	492	886
	long vehicle	38	43	0	81		
Test	ship	70	144	26	240		
	car	18	1	0	19	135	279
	long vehicle	12	8	0	20		

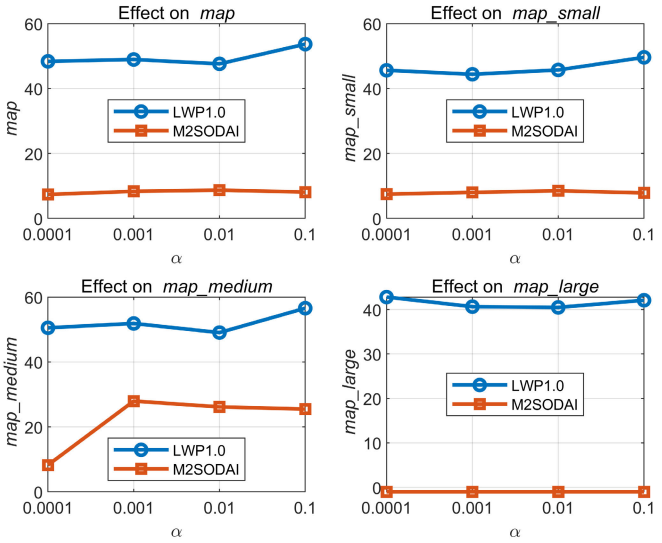


Fig. 4. Chart of detection performance variation based on the change of α.

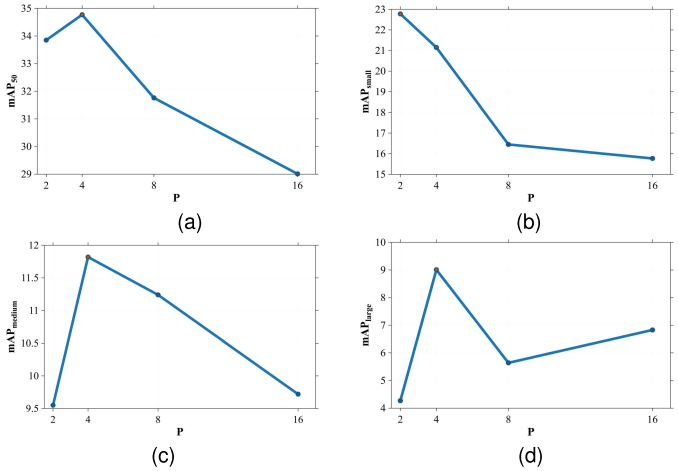


Fig. 5. Detection performance variation with respect to different values of P. (a) mAP_{50}. (b) mAP_{small}. (c) mAP_{medium}. (d) mAP_{large}.

All experiments were conducted on Ubuntu 20.04 with an Intel Core i9-9820X CPU, 64-GB RAM, and four NVIDIA GeForce RTX 2080 Ti GPUs (11-GB VRAM). All methods are implemented in PyTorch, with YOLO-based baselines using the Ultralytics framework and two-stage detectors following Detectron2-style implementations.

C. Parameter Determination

Since existing large-scale remote sensing HOD datasets are mainly distributed across the green, blue, red, and near-infrared spectral ranges, we heuristically set the parameter P to 4 during parameter configuration. Note that we also conduct relevant experiments to determine P, which will be discussed later in this section.

In the proposed method, the hyperparameter α in the SOPM plays a crucial role in performance. Thus, we iteratively adjust the value of α to find its optimal setting for the M2SODAI and LWP1.0 datasets. The experimental results are presented in Fig. 4.

As shown in Fig. 4, different datasets require different values of α, which reflects the different strengths of spectral-prior supervision needed by different object-size distributions. When α is set to 0.1, the proposed method achieves relatively better performance on the LWP1.0 dataset, especially in terms of mAP, mAP_{small}, and mAP_{medium}. This may be attributed to the fact that, although small and medium-sized objects still dominate LWP1.0, the proportion of large objects is higher than that in M2SODAI. Large objects usually contain more complex material compositions and provide richer spectral responses within the bounding boxes. Therefore, stronger spectral-prior supervision can be more beneficial for LWP1.0.

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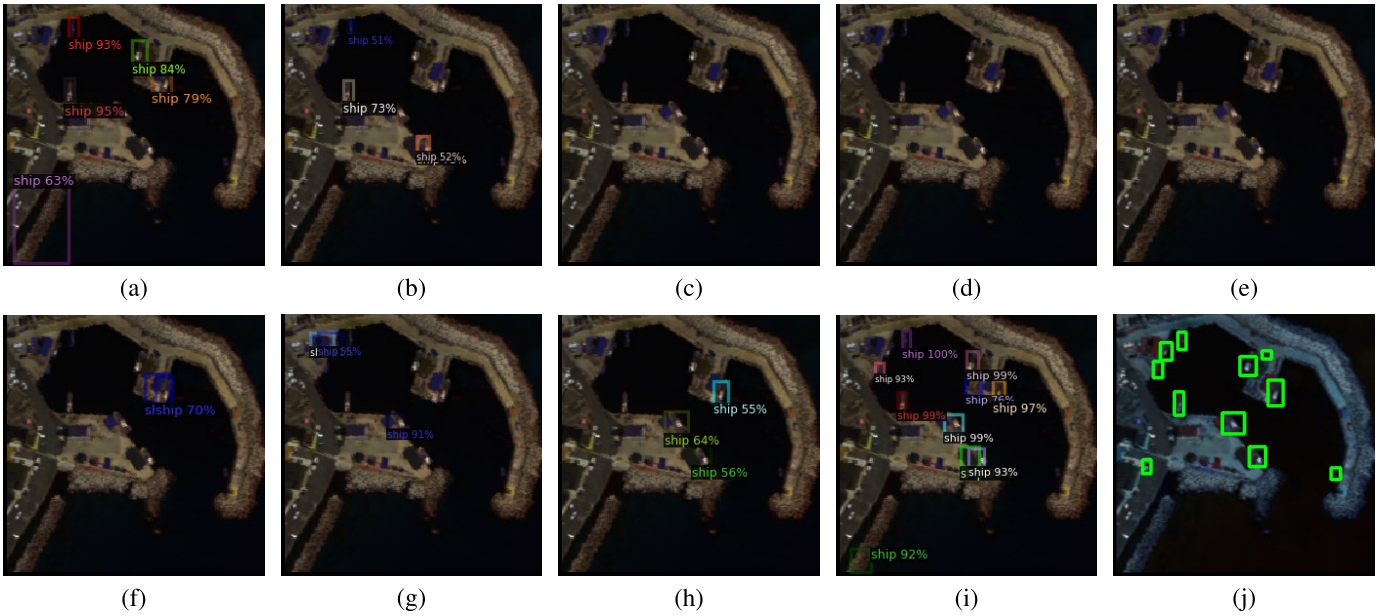


Fig. 6. Qualitative detection map of different methods in M2SODAI. (a) Faster-RCNN. (b) RT-DETRv2. (c) YOLOv5. (d) YOLOv8. (e) YOLOv10. (f) YOLO11. (g) Double-FPN. (h) S2A. (i) Ours. (j) Ground truth.

In contrast, M2SODAI is dominated by small objects, whose object regions contain less material information. As shown in Fig. 4, mAP and mAP_{small} achieve relatively better performance when $\alpha = 0.01$, whereas mAP_{medium} reaches its best value when $\alpha = 0.001$. Since small objects constitute the vast majority of M2SODAI, mAP and mAP_{small} are more representative for determining the parameter setting on this dataset. Therefore, $\alpha = 0.01$ is adopted for M2SODAI as a balanced choice, which preserves useful spectral guidance while avoiding excessive constraints imposed by SOPM on small-object-dominated scenes. Accordingly, α is set to 0.1 for LWP1.0 and LWP2.0, and to 0.01 for M2SODAI.

In addition, the parameter P in GLSEM plays a critical role in object–background discrimination. To this end, we empirically determine the value of P on the LWP2.0 dataset by adjusting it to 2, 4, 8, and 16. The corresponding experimental results are visualized in the form of figures, where four evaluation metrics mAP_{50}, mAP_{small}, mAP_{medium}, and mAP_{large} are adopted, as shown in Fig. 5.

As shown in Fig. 5, in small object detection, as P increases, more subtle spectral features can be captured. However, the spatial size of small objects is limited, and thus increasing P further may have a negative impact. In contrast, for the other three metrics, the detection performance exhibits an initial increase followed by a decrease as P grows. This phenomenon can be attributed to the fact that an excessive focus on local spectral differences may cause the model to overlook the overall spectral shape and fundamental material characteristics of the objects. Consequently, over-reliance on these fine details leads to a lack of global semantic understanding, diminishing the model's classification and detection capability. Nevertheless, $P = 4$ provides the best overall balance. Therefore, P is set to 4 in the subsequent experiments.

D. Comparison Results

In this section, we present a comprehensive analysis of the detection performance of different methods. Initially, the qualitative detection results, presented as detection maps, are shown in Fig. 6 for the M2SODAI dataset, Fig. 7 for the LWP1.0 dataset, and Fig. 8 for the LWP2.0 dataset. Subsequently, the quantitative evaluation results, measured by the mAP, are summarized in Tables III and IV.

In Fig. 6(a), Faster-RCNN shows limited detection performance. It fails to detect the two ships situated at the center of the HSI. Also, in the bottom-left corner, a strip-shaped land area is wrongly identified as a target, and this misidentification can be ascribed to the over-reliance of the model on the spatial features of the target. As shown in Fig. 6(b), RT-DETRv2 has limitations in the extraction of local features. This makes the detection of small objects, which depend heavily on detailed local information, challenging. In Fig. 6(c)–(f), YOLOv5, YOLOv8, YOLOv10, and YOLO11, as one-stage detectors, are unable to detect such small objects. This is especially obvious in cases where the spatial resolution is too low for the human eye to clearly distinguish the objects. Fig. 6(g) presents the detection map of Double-FPN. Although it is built upon a two-stage detector, the result is not satisfactory. The reason is that the PCA introduced disrupts the spectral features. As the first method specifically designed for HOD, S2A not only incorporates features obtained via PCA-based dimensionality reduction but also integrates a set of selected spectral band features, enabling it to achieve favorable performance within a single-stage detector framework.

In contrast, our method attains the best visual performance for small-sized objects. In such cases, spectral features function as the primary discriminative feature for differentiating between the background and the target. However, due to spectral variability, some nontargets are incorrectly detected, like the region in the bottom-left corner of the map. This issue requires further consideration in future research.

As depicted in Fig. 7(a), Faster-RCNN is unable to detect the prominent ship bow. Moreover, it misidentifies a region devoid of distinct ship features as a ship. Conversely, RT-DETRv2 encounters difficulties in detecting objects that lack comprehensive spatial features. This is because it is more

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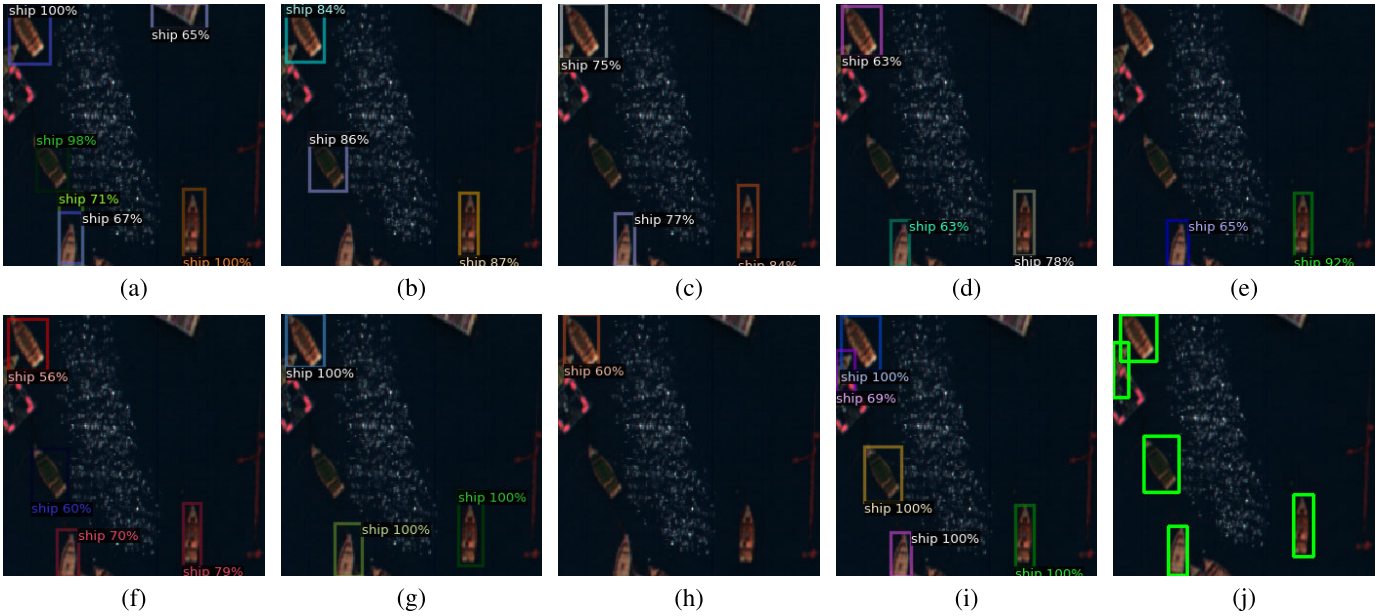


Fig. 7. Qualitative detection map of different methods in LWP1.0. (a) Faster-RCNN. (b) RT-DETRv2. (c) YOLOv5. (d) YOLOv8. (e) YOLOv10. (f) YOLO11. (g) Double-FPN. (h) S2A. (i) Ours. (j) Ground truth.

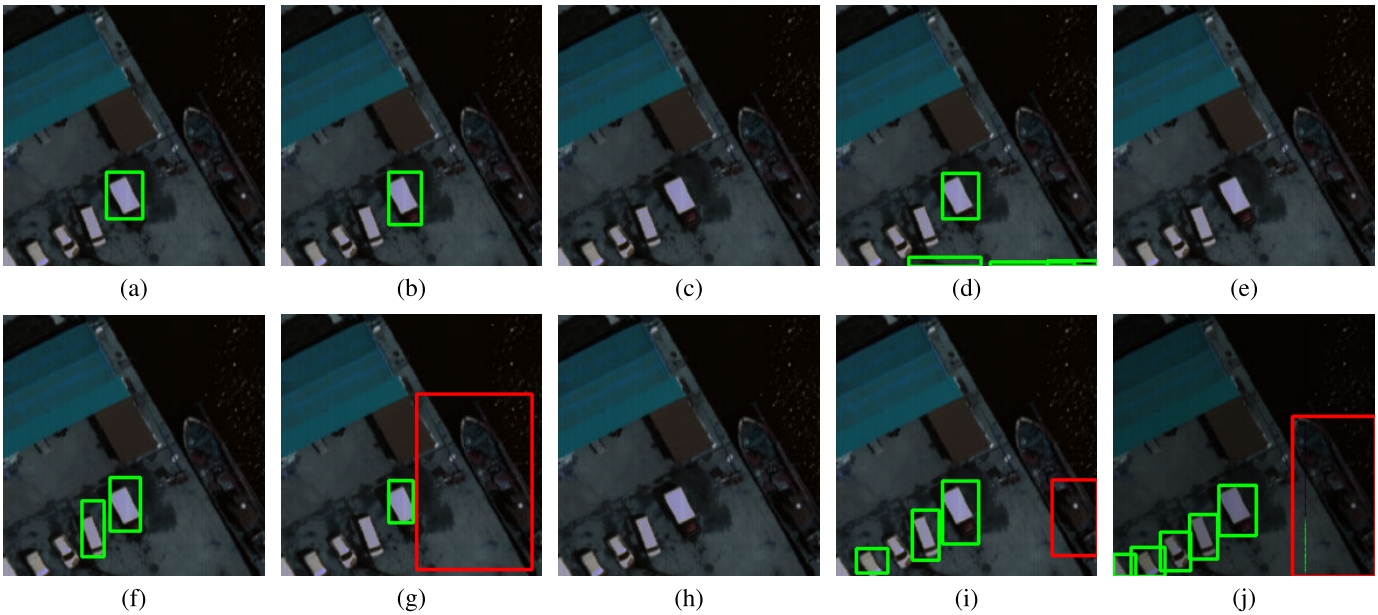


Fig. 8. Qualitative detection map of different methods in LWP2.0. (a) Faster-RCNN. (b) RT-DETRv2. (c) YOLOv5. (d) YOLOv8. (e) YOLOv10. (f) YOLO11. (g) Double-FPN. (h) S2A. (i) Ours. (j) Ground truth. The red box represents a ship, and the green box represents a long vehicle.

proficient at capturing global features but less effective in dealing with local details. YOLOv5, YOLOv8, YOLOv10, and YOLO11 all fail to detect even conspicuous objects. This is especially pronounced in the case of YOLOv10, which overlooks two targets positioned in the top-left corner and the central region of the HSI. We surmise that the root cause of this outcome is that single-stage models, designed for real-time processing, have limited capabilities in handling the complexity of HOD tasks. Double-FPN makes use of 30 bands generated by PCA to guide the HOD task. It fails to fully utilize the spectral features and low-level spatial features, thereby resulting in missing some targets. Although S2A further incorporates three additional spectral bands on this basis, these features are decoupled from the object detection task itself. As a result, it is difficult for the model to identify and exploit spectral bands that are truly beneficial for detection, which ultimately leads to suboptimal detection performance. In contrast to these methods, our approach is capable of capturing both spectral and spatial features. As a result, it can generate a robust visual detection map even when only partial spatial features are present.

In Fig. 8, the Faster-RCNN shows a similar detection result with RT-DETRv2 and YOLOv8, where only a long vehicle is detected. In YOLOv5 and YOLOv10, the confidence of the object may be so low that many objects are missed. YOLO11 is able to detect the other long vehicle. However, no evidence

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TABLE III
QUANTITATIVE EXPERIMENTAL RESULTS OF COMPARATIVE METHODS IN M2SODAI AND LWP1.0

M2SODAI						
	mAP	mAP_{50}	mAP_{75}	mAP_{small}	mAP_{medium}	mAP_{large}
Faster-RCNN	4.35	17.14	**3.59**	4.35	8.40	N/A
RT-DETRv2	4.79	16.62	1.24	4.86	5.17	N/A
YOLOv5	4.10	16.95	0.36	4.10	8.09	N/A
YOLOv8	4.55	17.08	0.61	4.41	12.96	N/A
YOLOv10	5.02	18.73	0.16	4.92	12.51	N/A
YOLO11	3.78	15.27	0.24	3.81	3.81	N/A
DoubleFPN	6.71	22.72	1.84	6.26	**31.26**	N/A
S2A	7.95	28.42	2.44	8.42	2.42	N/A
Ours	**8.69**	**29.20**	2.05	**8.49**	26.16	N/A
LWP1.0						
	mAP	mAP_{50}	mAP_{75}	mAP_{small}	mAP_{medium}	mAP_{large}
Faster-RCNN	25.53	49.25	24.90	23.57	24.96	54.37
RT-DETRv2	32.42	57.84	33.16	27.81	33.01	60.64
YOLOv5	21.16	43.36	15.08	14.78	24.33	37.02
YOLOv8	42.20	79.21	42.32	37.39	44.95	53.54
YOLOv10	51.36	82.29	58.44	44.40	54.31	**73.29**
YOLO11	36.69	73.89	30.31	31.95	39.58	36.40
DoubleFPN	16.42	42.18	9.16	12.87	18.76	13.82
S2A	19.47	44.99	13.53	8.07	25.52	3.38
Ours	**53.68**	**83.79**	**62.20**	**49.61**	**56.57**	42.08

TABLE IV
QUANTITATIVE EXPERIMENTAL RESULTS OF COMPARATIVE METHODS IN LWP2.0

LWP2.0					
	mAP	mAP_{50}	AP_{ship}	AP_{car}	$AP_{long\ vehicle}$
Faster-RCNN	6.71	14.40	19.32	17.11	6.78
RT-DETRv2	11.44	27.87	22.68	32.50	28.43
YOLOv5	2.78	10.23	12.57	8.31	9.81
YOLOv8	2.37	6.62	9.50	10.07	0.28
YOLOv10	10.14	22.11	8.49	33.66	24.18
YOLO11	3.91	10.39	8.11	9.56	13.51
DoubleFPN	5.89	16.35	18.81	7.64	22.60
S2A	3.63	12.27	4.70	10.54	21.57
Ours	**17.02**	**34.77**	**24.28**	**45.39**	**34.65**

of a ship was detected. In contrast, Double-FPN effectively identified the ship's outline but exhibited limited performance in detecting the other objects. Although the spectral curves of ships and background exhibit clear differences in the original hyperspectral space, S2A fails to exploit this discriminability effectively. This is because the discriminative spectral cues are largely diluted or discarded during PCA-based dimensionality reduction and static band selection, which are not optimized for object-level detection. As a result, the spectral separability observed in the raw data cannot be reliably preserved in the feature space used by S2A, leading to detection failure.

In this scene, our approach correctly identifies most objects, with only one long vehicle missed. The long vehicle object may receive a lower score and therefore not be recognized due to its spectral characteristics being similar to those of surrounding long vehicles. Due to the ship's large size and the presence of diverse materials, the model tended to focus on specific spectral features (such as those in the middle section of the hull) when guided by spectral information. As a result, it failed to accurately delineate the entire ship, which highlights a limitation that warrants further investigation in future work.

We note that qualitative visualizations are generated at a fixed confidence threshold, whereas mAP integrates precision–recall over varying thresholds; therefore, the visual impression under a particular threshold may not perfectly align with the overall mAP. The corresponding quantitative *mAP* indices are summarized in Tables III and IV.

In the M2SODAI dataset, small ships represent the predominant object category. Due to its superior performance in detecting small objects, our method attains the best overall

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(a) (b)

Fig. 9. Qualitative detection maps obtained by reducing the confidence threshold to 0.1. (a) YOLOv10. (b) YOLO11.

mAP, mAP_{50}, and mAP_{small}, while its mAP_{75} is not the best, indicating that the proposed spectral guidance is more beneficial to object recall and small-object detection than highly precise localization. Our method faces difficulties in detecting medium-sized objects. The root cause of this issue has been comprehensively explored in the ablation study section. Despite these challenges, our method still ranks second in terms of mAP_{medium}. It should be noted that the *mAP* for large objects is set as N/A because the M2SODAI test dataset does not contain any large objects.

In the LWP1.0 dataset, the proposed method achieves the best overall performance in terms of *mAP*, mAP_{50}, mAP_{75}, mAP_{small}, and mAP_{medium}, indicating its strong effectiveness on the dominant small- and medium-sized objects. However, the advantage does not extend to mAP_{large}. This result should be interpreted with caution, because the number of large objects in the LWP1.0 test set is very limited, making the corresponding evaluation more susceptible to statistical fluctuation.

In the LWP2.0 dataset, our method achieves superior performance in detecting various types of objects, resulting in the highest *mAP* and mAP_{50} among all compared approaches.

A crucial question emerges: why does YOLOv10 outperform YOLO11 quantitatively in terms of *mAP*, whereas the qualitative visualization appears to show the opposite trend in some cases? This discrepancy is mainly caused by the difference between threshold-free quantitative evaluation and fixed-threshold visualization. Specifically, *mAP* is computed by ranking detections according to their confidence scores and integrating the precision–recall curve over multiple operating points and IoU thresholds. Therefore, it reflects the overall confidence ranking, recall ability, and localization accuracy of a detector. In contrast, the qualitative visualization only displays detections whose confidence scores are higher than a fixed threshold, i.e., 0.5 in our original visualization. As a result, low-confidence but correctly localized detections may be suppressed in the visual results.

To further verify this explanation, we conduct an additional threshold analysis on the LWP1.0 dataset using YOLOv10 and YOLO11 as examples. When the confidence threshold is reduced to 0.1, as shown in Fig. 9, more low-confidence detections are retained. The visualization shows that YOLOv10 can produce more candidate boxes around object regions, and some of these boxes have better localization consistency with the objects, although their confidence scores are relatively low. This advantage is captured by the *mAP* calculation because *mAP* evaluates detections over the full confidence-ranked list and across IoU thresholds. Therefore, YOLOv10 can achieve a higher quantitative *mAP* even though YOLO11 may look visually cleaner under a single fixed confidence threshold.

E. Ablation Study

To verify the efficacy of the proposed modules, we perform an ablation study on both the GLSEM and SOPM components in the LWP1.0 dataset and the M2SODAI dataset.

The ablation results on the LWP1.0 dataset are shown at the top of Table V. When the GLSEM is removed, there is a significant degradation in the performance for detecting small and medium-sized objects. Conversely, the detection performance for large objects improves. This clearly reveals that the GLSEM is crucial for enhancing the accuracy of detecting small and medium-sized objects. The reasons behind the observed increase in mAP_{large} are further discussed in the qualitative analysis section. When the SOPM is removed, there is a consistent decrease across all *mAP* metrics. This demonstrates the vital role and efficacy of the SOPM in enhancing the performance of the proposed method. When both SOPM and GLSEM are removed, the detection performance for small and medium-sized objects drops substantially, which is particularly important because these objects dominate the LWP1.0 dataset. In contrast, the model achieves better detection performance for large objects, thereby leading to an increase in mAP_{50}. For large objects, a single bounding box may contain multiple materials, resulting in more heterogeneous spectral responses. This may weaken the effectiveness of GLSEM and SOPM, since both modules mainly process spectral information. Therefore, these two modules are more beneficial for small objects, while their contribution to large-object detection is relatively limited.

The ablation results on the M2SODAI dataset are reported at the bottom of Table V. When either GLSEM or SOPM is removed individually, the detection performance for small and medium-sized objects degrades to varying degrees. Since large-sized objects are absent in this dataset, the corresponding results are denoted as N/A. When both modules are removed, a similar phenomenon to that observed on the LWP1.0 dataset occurs: the performance for relatively larger objects, such as medium-sized objects, is still improved. This improvement can be attributed to the same reason discussed above.

Meanwhile, an increase in mAP_{75} is also observed. Since mAP_{75} is evaluated at an IoU threshold of 75%, it reflects a stricter requirement for localization accuracy. This indicates that the two modules may not be beneficial for highly precise localization in this case. Different from the LWP1.0 dataset, the M2SODAI dataset is dominated by small-sized objects. Although the two modules enhance the modeling of spectral features, a higher IoU threshold imposes more stringent requirements on spatial localization, especially when small objects account for the majority of the dataset. Therefore, after removing the two modules, mAP_{75} instead shows an improvement on the M2SODAI dataset.

To further investigate the role of GLSEM, we conduct a qualitative ablation study. Specifically, we remove GLSEM from the framework and adopt gradient-weighted class activation mapping (Grad-CAM) [49] to visualize the response maps. The comparative visualization results are presented in Fig. 10(a) and (b). As observed, the model equipped with GLSEM exhibits a stronger capability to capture fine-grained

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TABLE V
QUANTITATIVE ABLATION EXPERIMENTAL RESULTS ON LWP1.0 AND M2SODAI DATASETS

LWP1.0						
	mAP	mAP_{50}	mAP_{75}	mAP_{small}	mAP_{medium}	mAP_{large}
original one	**53.681**	83.792	**62.196**	**49.615**	**56.573**	42.080
remove GLSEM	50.815	82.409	57.938	45.472	53.275	65.639
remove SOPM	48.091	77.447	55.813	36.712	54.131	39.125
remove GLSEM+SOPM	50.896	**84.388**	58.356	42.562	54.096	**67.537**
M2SODAI						
	mAP	mAP_{50}	mAP_{75}	mAP_{small}	mAP_{medium}	mAP_{large}
original one	**8.686**	**29.203**	2.054	**8.493**	26.164	N/A
remove GLSEM	7.871	26.620	1.657	7.845	14.367	N/A
remove SOPM	7.706	27.903	1.956	7.588	20.309	N/A
remove GLSEM+SOPM	7.791	28.038	**2.215**	7.487	**28.997**	N/A

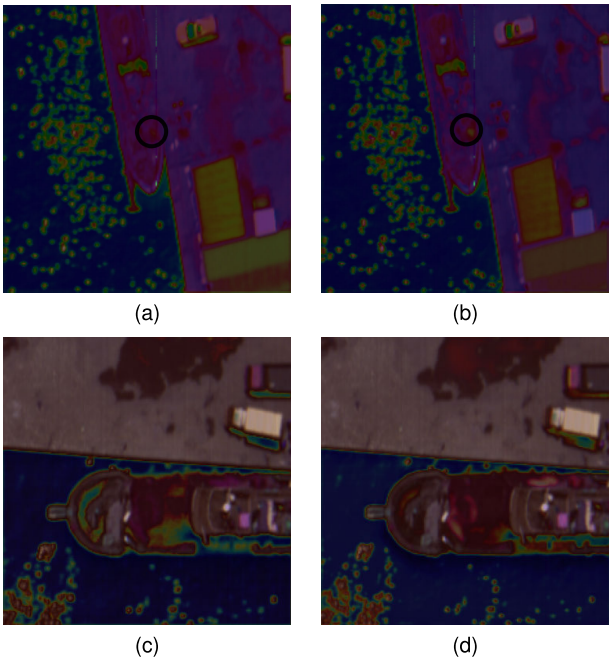


Fig. 10. Qualitative visualization results for the ablation analysis of GLSEM and SOPM. (a) Grad-CAM response map without GLSEM. (b) Grad-CAM response map with GLSEM. (c) Feature response map of the third spectral group ($\mathbf{X}_3$). (d) SOPM-enhanced objectness response map.

feature details, particularly in the regions highlighted by black circles. In these areas, GLSEM enhances the discrimination of subtle material-related characteristics on ships, which indirectly validates its effectiveness in extracting detailed spectral information. The effect of GLSEM on large objects appears less stable than that on small and medium objects.

Since the final step of GLSEM lacks convolutional operations, the gradient limitations of Grad-CAM prevent us from using the last variable generated by GLSEM. To evaluate SOPM's effectiveness, we visualize the response map of the third group ($\mathbf{X}3$) and compare it with the response map after SOPM, as shown in Fig. 10(c) and (d). Our analysis reveals that the SOPM-generated response map shows smoother feature targets compared to $\mathbf{X}_3$ in GLSEM. This improvement is due to the inclusion of learned spectral prior features, which help SOPM focus the model on objects, thus enhancing overall performance.

TABLE VI
COMPUTATIONAL COST COMPARISON AVERAGED OVER 50 VALIDATION IMAGES

Variant	Params (M)	FLOPs (G)	Time (ms/img)
Original	123.1701	253.8734	42.5041
w/o GLSEM	122.5680	194.6089	28.6543
w/o SOPM	122.5520	230.0131	39.5673

Furthermore, to analyze the function of GLSEM, we select a scene containing objects with identical material properties but different semantic categories. Specifically, two representative points corresponding to a wooden harbor trestle and a wooden ship are selected, as illustrated in Fig. 11, marked in red and green, respectively.

SAM is employed for quantitative evaluation. SAM measures the spectral angular difference among spectral signatures and reflects the degree of spectral modulation after the GLSEM. A larger SAM value indicates increased spectral separability between objects with similar materials.

By computing the SAM between the original and GLSEM-enhanced spectral curves, we qualitatively analyze the spectral variation introduced by GLSEM, as shown in Fig. 11(j). When $P = 4$, the spectral discrepancy between the two objects is moderately enlarged compared to the original spectra, while the corresponding mAP_{50} simultaneously reaches (see Fig. 5) its maximum among all parameter settings. In contrast, both smaller and larger values of P lead to inferior detection performance.

As shown in Fig. 11(j), when P = 2, the local spectral modulation is relatively weak, although the spectral difference is already large. When P = 4, the modulation becomes more effective, leading to better detection performance even though the spectral difference is not the largest. These observations indicate that GLSEM does not simply amplify material information, but selectively enhances discriminative spectral cues while maintaining stable object-level semantic representation.

GLSEM and SOPM introduce additional computational overhead and parameters, which inevitably increase the inference time. Therefore, Table VI reports the number of

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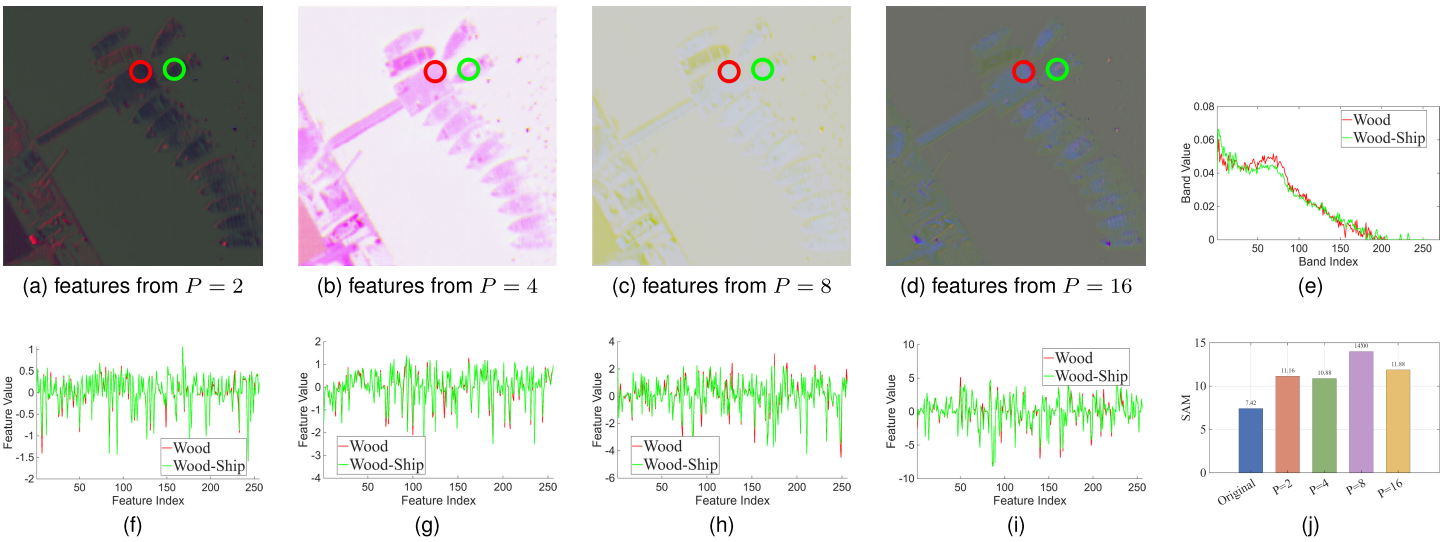


Fig. 11. (a)–(d) Features generated with different P values, where the 30th, 80th, and 170th channels are selected for pseudo-color visualization. (e) Spectral curves of selected pixels from the original HSI. (f)–(i) Modulated spectral curves, which correspond one-to-one with the cases (a)–(d). (j) SAM values under different P settings and the original HSI. Red and green circles denote the selected wooden harbor trestle and wooden ship pixels, respectively.

parameters, FLOPs, and inference time of the original method and its variants with the proposed modules removed. It is worth noting that GLSEM only consists of several parallel 3×3 convolutional layers; thus, it increases the number of parameters by only 0.6021 M. The full model requires 253.87 GFLOPs, while the variants without GLSEM and SOPM require 194.61 and 230.01 GFLOPs, respectively. This indicates that the proposed modules introduce limited additional computation, and the model still achieves efficient inference with an average time of 42.50 ms/image. The full model requires 42.50 ms/image for inference, which is only moderately higher than the ablated variants. This shows that the proposed GLSEM and SOPM modules introduce acceptable runtime overhead while maintaining efficient inference.

In summary, compared with the original spectra, GLSEM enhances object–background separability within a moderate range. Greater spectral distinction is not necessarily better, because effective detection requires stronger discriminative cues while still preserving the basic material characteristics and stable object-level semantic structure. Guided by SOPM, the detector can identify regions whose spectral responses are similar to the prior spectral signatures, helping the network focus on object-like areas and introducing additional discriminative information.

V. Conclusion

This article addresses two major limitations in current HOD methods: objects may share similar materials with the background, making them more difficult to distinguish, and the underuse of spectral features. To solve these, we propose SPG-OD. First, we introduce GLSEM to enhance discriminative local spectral cues, improving object–background discrimination. We also design SOPM to learn prior object spectral representations, enhancing spectral feature representation. Extensive experiments show that our method achieves competitive performance.

Although the proposed method achieves competitive overall performance, it still has several limitations. First, its advantages are more evident for small than for larger objects. Larger objects usually contain multiple materials and more complex spatial structures, and their spectral responses may vary across different object parts. In this case, the spectral-prior guidance may emphasize locally discriminative regions rather than uniformly covering the entire object extent, leading to less stable localization for large objects. Second, false responses may occur in background regions with spectral characteristics similar to real objects. This is because the current framework mainly exploits spectral cues for object–background discrimination, while some background materials may also show object-like spectral responses. In such cases, the model may assign high responses to these regions even though they do not form complete objects.

Therefore, future work will further combine object-level localization with pixel-level spectral interpretation, so that the model can not only detect what the object is but also analyze its internal material or component composition. This may help distinguish true objects from spectrally similar background regions and provide more informative HSI interpretation.

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